

# Mask-Path InfoNCE without EMA in a Global Image-Level JEPA: A CIFAR Study

Katakis Sofoklis  
kat.sofoklis@gmail.com

## Abstract

Joint-Embedding Predictive Architectures (JEPAs) commonly stabilize latent prediction with an exponential-moving-average (EMA) target encoder. We test whether applying InfoNCE directly to the masked predictive pair can instead train a single ResNet-18 encoder with stop-gradient targets and no EMA. Under a matched 100-epoch CIFAR-10 protocol across three seeds, mask-path InfoNCE reaches  $42.6 \pm 3.6\%$   $k$ -nearest-neighbor accuracy and  $43.0 \pm 3.4\%$  linear-probe accuracy, compared with  $41.6 \pm 2.9\%$  and  $41.3 \pm 2.7\%$  for cosine JEPA with an EMA target. The no-EMA method has slightly higher means, although the observed variability does not establish a decisive advantage. These results support mask-path contrast as a workable no-EMA option only in this restricted CIFAR setting.

## 1 Introduction

Self-supervised learning (SSL) for visual representations spans two influential families: *contrastive* instance discrimination [4, 8] and *predictive* joint embedding (JEPA) [1, 10]. I-JEPA predicts target-block embeddings from a context encoder and uses regression in latent space with an EMA target network. C-JEPA [11] combines I-JEPA with variance-invariance-covariance regularization (VICReg) [3] to stabilize training and improve convergence on ImageNet. These methods suggest several plausible substitutes for an EMA target: batch negatives, variance-covariance constraints, or distribution matching. However, a JEPA pipeline with both masked and augmented views introduces a placement choice: the anti-collapse signal can act on masked prediction or on a separate augmentation path.

We therefore ask: *for a single-encoder JEPA without EMA, should contrast act directly on masked*

*prediction, or should masked prediction remain non-contrastive while a separate augmentation path prevents collapse?* We compare cosine mask→clean prediction plus augmentation→clean InfoNCE with mask-path InfoNCE and non-contrastive augmentation regularizers.

**Contributions.** (i) A controlled formulation that separates the masked predictive path from the augmentation anti-collapse path; (ii) a matched CIFAR-10 comparison of contrastive, variance-covariance, and Gaussian regularization with a single ResNet-18 encoder; (iii) evidence that mask-path InfoNCE is the strongest tested no-EMA objective, while augmentation-side regularization does not improve on that baseline.

## 2 Related work

**Predictive JEPA.** I-JEPA [1] predicts latent targets under masking with smooth-L1 / L2 and an EMA teacher, following the JEPA agenda of LeCun [10].

**C-JEPA.** C-JEPA [11] argues that EMA alone is insufficient to prevent collapse and that mean-patch prediction is underconstrained; it augments I-JEPA with VICReg [3] variance / covariance / invariance terms and reports faster ImageNet convergence. Our note shares C-JEPA’s goal of combining predictive and contrastive-style signals, but studies a much smaller CIFAR recipe: one encoder, stop-grad (no EMA), cosine JEPA on the mask path, and either augmentation InfoNCE or a lightweight regularizer (mean squared error with variance and covariance penalties, VICReg, or Sketched Isotropic Gaussian Regularization (SIGReg) [2]).

**Contrastive and non-contrastive SSL.** SimCLR [4] and MoCo [8] use InfoNCE on two augmented views. BYOL [6] and SimSiam [5] avoid ex-

explicit negatives via predictors and stop-grad / EMA. Barlow Twins [12] and VICReg [3] prevent collapse with redundancy-reduction / variance-covariance penalties rather than negatives.

### 3 Methods and Objectives

**Views and notation.** For each image  $x$ , we form a lightly augmented clean view  $x_c$ , a block-masked view  $x_m$ , and a strongly augmented view  $x_a$ . A single encoder  $f_\theta$  maps each view to a 512-dimensional representation, and a predictor  $g_\phi$  is used only on the masked path:

$$z_c = f_\theta(x_c), \quad \hat{z}_m = g_\phi(f_\theta(x_m)), \quad z_a = f_\theta(x_a). \quad (1)$$

The predictor is discarded at evaluation. Except for the EMA reference, all objectives use the same online encoder. The notation  $\text{sg}(\cdot)$  denotes stop-gradient. It is applied to the clean target in the cosine and InfoNCE predictive pairs, but not between the two branches of the variance-covariance or SIGReg augmentation regularizers.

**Cosine JEPA.** The non-contrastive masked-prediction objective is

$$L_{\cos} = 1 - \frac{\hat{z}_m^\top \text{sg}(z_c)}{\|\hat{z}_m\|_2 \|\text{sg}(z_c)\|_2}. \quad (2)$$

Averaging over the batch is implicit. This objective enforces invariance between masked and clean views but contains no explicit term requiring features from different images to remain distinct. Consequently, a single-encoder version can admit collapsed or weak representations. The EMA baseline uses the same cosine loss but computes the clean target with a momentum encoder  $f_\xi$ , updated as  $\xi \leftarrow m\xi + (1-m)\theta$  with  $m = 0.996$ .

**InfoNCE.** For query-key pairs  $q_i, k_i$  in a batch of size  $B$ , we use

$$L_{\text{NCE}}(q, k) = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\bar{q}_i^\top \bar{k}_i / \tau)}{\sum_{j=1}^B \exp(\bar{q}_i^\top \bar{k}_j / \tau)}, \quad (3)$$

where bars denote  $\ell_2$  normalization and  $\tau = 0.1$ . The matching view of the same image is the positive; the other  $B - 1$  clean embeddings are negatives. Stop-gradient is applied to every key. We compare two placements:

$$L_{\text{mask-NCE}} = L_{\text{NCE}}(\hat{z}_m, \text{sg}(z_c)), \quad (4)$$

$$L_{\cos+\text{aug-NCE}} = L_{\cos} + L_{\text{NCE}}(z_a, \text{sg}(z_c)). \quad (5)$$

Thus, mask-only InfoNCE makes the predictive task itself contrastive, whereas the second objective preserves non-contrastive masked prediction and places instance discrimination only on the augmentation path. Both active terms have unit weight.

**MSE, variance, and covariance.** To test negative-free anti-collapse regularization, we replace augmentation InfoNCE with the three VICReg building blocks. For  $z \in \mathbb{R}^{B \times D}$ ,

$$L_{\text{inv}} = \frac{1}{BD} \|z_a - z_c\|_F^2, \quad (6)$$

$$v(z) = \frac{1}{D} \sum_{d=1}^D \max\left(0, 1 - \sqrt{\text{Var}(z_{:,d}) + \epsilon}\right), \quad (7)$$

$$c(z) = \frac{1}{D} \sum_{r \neq s} \text{Cov}(z)_{rs}^2, \quad (8)$$

with  $\epsilon = 10^{-4}$ . The invariance term aligns the two views, the variance hinge prevents individual dimensions from becoming constant, and the covariance penalty discourages redundant dimensions. The complete objective is

$$L_{\text{MVC}} = L_{\cos} + \lambda_i L_{\text{inv}} + \lambda_v [v(z_a) + v(z_c)] + \lambda_c [c(z_a) + c(z_c)]. \quad (9)$$

We evaluate unit weights  $(\lambda_i, \lambda_v, \lambda_c) = (1, 1, 1)$  and the VICReg-scale setting (25, 25, 25) reported as separate rows. Gradients flow through both  $z_a$  and  $z_c$  in these three terms.

**SIGReg.** Sketched Isotropic Gaussian Regularization (SIGReg) replaces the explicit variance and covariance penalties with a distribution-matching term. We concatenate both augmentation branches,  $Z = [z_a; z_c]$ , project  $Z$  onto 256 deterministically seeded random unit directions, and compare each projected empirical characteristic function with that of  $\mathcal{N}(0, 1)$  using the Epps-Pulley statistic [2]. In compact form,

$$L_{\text{SIG}} = L_{\cos} + L_{\text{inv}} + \frac{1}{R} \sum_{r=1}^R \mathcal{E}(Zu_r, \mathcal{N}(0, 1)), \quad R = 256, \quad (10)$$

where  $u_r$  is a random unit direction and  $\mathcal{E}$  is the weighted characteristic-function discrepancy. All three terms use unit weight. SIGReg encourages a non-degenerate, approximately isotropic latent distribution while  $L_{\text{inv}}$  preserves augmentation invariance.

Table 1: CIFAR-10, ResNet-18, evaluated at the fixed 100-epoch endpoint. Learned-model results are mean  $\pm$  sample standard deviation over seeds 42, 7, and 123. Random chance is included as a reference baseline. Cosine JEPA + augmentation InfoNCE places batch negatives only on the augmentation path. Hybrids = cosine JEPA (mask) + listed augmentation/clean regularizer (no EMA). All reported runs are complete.

| Method                                      | $k$ -NN                             | Linear                              | Seeds |
|---------------------------------------------|-------------------------------------|-------------------------------------|-------|
| <i>Reference baselines</i>                  |                                     |                                     |       |
| Random classifier (chance)                  | 0.100                               | 0.100                               | –     |
| Supervised CE                               | $0.949 \pm 0.001$                   | $0.948 \pm 0.001$                   | 3     |
| <i>Self-supervised learning</i>             |                                     |                                     |       |
| EMA JEPA cosine                             | $0.416 \pm 0.029$                   | $0.413 \pm 0.027$                   | 3     |
| JEPA InfoNCE (mask only; no EMA)            | <b><math>0.426 \pm 0.036</math></b> | <b><math>0.430 \pm 0.034</math></b> | 3     |
| Cosine JEPA + augmentation InfoNCE (no EMA) | $0.402 \pm 0.010$                   | $0.399 \pm 0.019$                   | 3     |
| JEPA + MSE/variance/covariance (no EMA)     | $0.397 \pm 0.002$                   | $0.346 \pm 0.063$                   | 3     |
| JEPA + VICReg (no EMA; 25/25/25)            | $0.374 \pm 0.004$                   | $0.393 \pm 0.007$                   | 3     |
| JEPA + SIGReg (no EMA)                      | $0.383 \pm 0.008$                   | $0.407 \pm 0.004$                   | 3     |

Table 2: CIFAR-100, ResNet-18, seed 42, evaluated at the fixed 100-epoch endpoint. These exploratory single-seed results are reported separately from the three-seed CIFAR-10 comparison and do not include uncertainty estimates.

| Method                                      | $k$ -NN      | Linear       | Epoch |
|---------------------------------------------|--------------|--------------|-------|
| Random classifier (chance)                  | 0.010        | 0.010        | –     |
| EMA JEPA cosine                             | 0.125        | 0.129        | 100   |
| JEPA InfoNCE (mask only; no EMA)            | <b>0.155</b> | <b>0.150</b> | 100   |
| Cosine JEPA + augmentation InfoNCE (no EMA) | 0.118        | 0.106        | 100   |

**Controlled comparison.** Across the no-EMA rows, the encoder, predictor, views, optimizer, and training budget are fixed; only the objective and its stated weights change. This separation lets us test whether anti-collapse behavior depends not only on the regularizer family but also on the path to which it is applied.

## 4 Experimental protocol

**Data and architecture.** CIFAR-10 [9]; ResNet-18 [7] with embedding dimension 512; a predictor multilayer perceptron (MLP) on the mask path. Block masking with mask ratio 0.6. Strong augs: random resized crop, horizontal flip, color jitter ( $B=C=S=0.4$ , hue 0.1).

**Optimization.** Adam, learning rate  $3 \times 10^{-4}$ , batch size 256, 100 epochs, InfoNCE temperature  $\tau=0.1$ , one encoder (no EMA).

**Evaluation.** Frozen encoder;  $k$ -NN ( $k=20$ ) and linear probe (standard CIFAR split). Evaluate every 10 epochs and select the checkpoint with best  $k$ -NN.

**Hardware and runtime.** Experiments were run on NVIDIA GeForce RTX 5060 Ti GPUs (16 GB), using one GPU per run. Including periodic frozen-feature evaluation, the 100-epoch mask-only InfoNCE run took 11,721 s (3.26 h), while cosine JEPA plus augmentation InfoNCE took 10,546 s (2.93 h). The two runs can be executed concurrently on two GPUs, for approximately 3.26 h wall-clock time.

**Baselines (same stack).**

- Supervised cross-entropy (CE; upper bound);
- EMA JEPA cosine (teacher EMA);
- JEPA InfoNCE on the mask path only (no augmentation term);
- Cosine JEPA + augmentation InfoNCE;
- Hybrids: cosine JEPA (mask→clean, no EMA) + MSE/variance/covariance, SIGReg [2], or VICReg [3] on augmentation/clean;

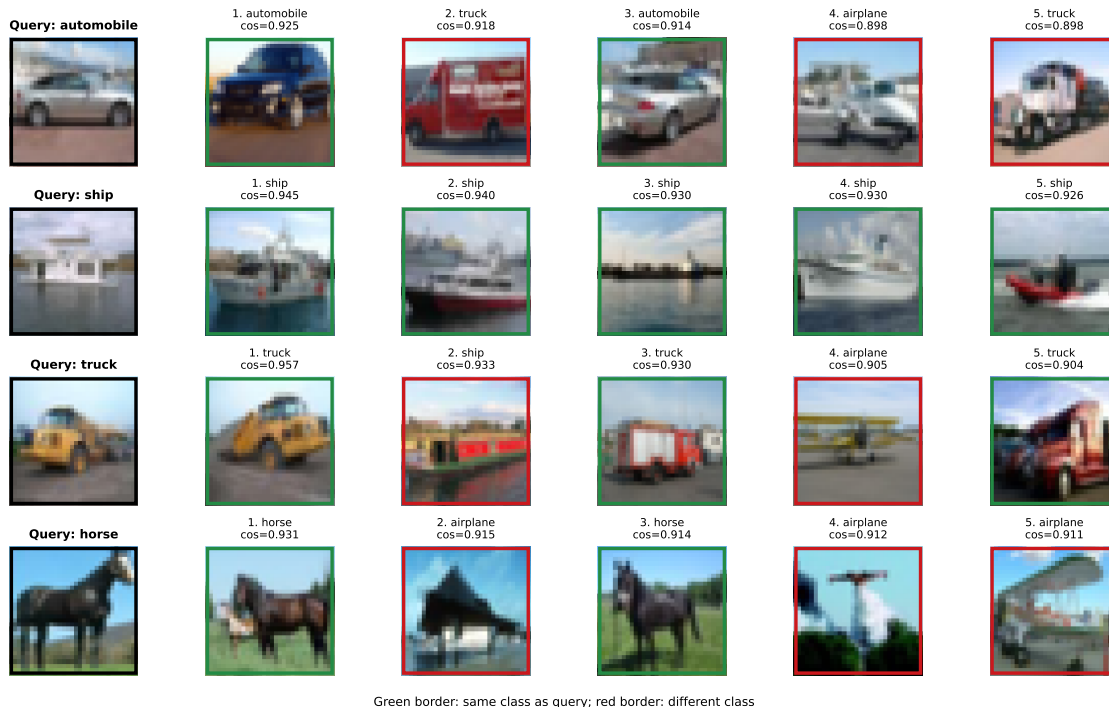


Figure 1: Five nearest training images for four test queries under cosine similarity in the frozen mask-only InfoNCE representation. The encoder is the best checkpoint at epoch 80. The four representative queries span automobile, ship, truck, and horse; their fixed test indices are provided with the figure metadata. Retrieval uses all 50,000 training images with deterministic evaluation preprocessing. Green borders indicate the query class and red borders a different class. The examples expose both coherent neighborhoods and cross-class confusion, consistent with the moderate aggregate  $k$ -NN accuracy.

## 5 Results

All numbers below are CIFAR-10, ResNet-18 ( $D=512$ ), batch 256, 100 epochs, frozen  $k$ -NN ( $k=20$ ) and linear probe. We report the fixed epoch-100 endpoint as mean  $\pm$  sample standard deviation over three seeds (42, 7, and 123) for both supervised and self-supervised models.

**Qualitative neighborhoods.** Figure 1 shows the five most similar training images for fixed test queries spanning four classes. The class-consistent ship neighborhood contrasts with mixed automobile, truck, and horse neighborhoods. This qualitative variation helps explain why local similarity is useful but far from class-perfect.

**Mask-path contrast is strongest.** Mask-only InfoNCE is the best tested SSL method by both  $k$ -NN ( $0.426 \pm 0.036$ ) and linear probe ( $0.430 \pm 0.034$ ). It exceeds the EMA cosine-JEPA reference by about

one percentage point on each mean while using no teacher encoder. This suggests that batch negatives on the predictive pair can supply an effective anti-collapse signal in the present regime, although the overlapping uncertainty cautions against claiming a decisive advantage over EMA.

**Augmentation-side contrast is not an equivalent substitute.** Cosine JEPA plus augmentation InfoNCE reaches  $0.402 \pm 0.010$   $k$ -NN and  $0.399 \pm 0.019$  linear accuracy, trailing mask-only InfoNCE on both means. Its smaller variance does not recover the average representation quality, supporting the view that contrast placement matters.

**Distributional regularizers trail mask-path contrast.** The cosine-JEPA regularizers do not recover the mask-only InfoNCE result. MSE/variance/covariance has stable  $k$ -NN ( $0.397 \pm 0.002$ ) but a variable linear probe ( $0.346 \pm 0.063$ ). VICReg reaches

$0.374 \pm 0.004 / 0.393 \pm 0.007$ , while SIGReg reaches  $0.383 \pm 0.008 / 0.407 \pm 0.004$  ( $k$ -NN/linear). The mismatch between neighborhood and linear metrics cautions against ranking representations with only one evaluator.

## 6 Discussion and limitations

**Relation to C-JEPA.** C-JEPA [11] combines I-JEPA with VICReg on large-scale ImageNet pre-training and retains an EMA teacher. We target a minimal CIFAR recipe without EMA: cosine JEPA on the mask path for “ours” and for the regularizer hybrids (MSE/variance/covariance, SIGReg [2], VICReg [3]).

**Limitations.** CIFAR-10, ResNet-18, and three seeds are insufficient for broad claims about JEPA training, and the objectives were not tuned with equal hyperparameter budgets. We lack a matched augmentation-InfoNCE-only control and ImageNet-scale validation. Although fixed-epoch reporting avoids selecting the endpoint by test  $k$ -NN, hyperparameter development still used the same benchmark; future experiments should use a held-out validation split. Finally, our global image-level target simplifies I-JEPA’s block-level prediction task.

**Takeaway.** A single encoder can learn non-collapsed masked representations without EMA, but the successful ingredient is not merely the presence of an anti-collapse loss. At epoch 100, mask-path InfoNCE is the strongest tested choice on average ( $0.426 \pm 0.036$   $k$ -NN,  $0.430 \pm 0.034$  linear); cosine prediction paired with augmentation-side InfoNCE or distributional regularization is weaker under the evaluated settings. The practical lesson is to treat loss placement as a first-class design choice in predictive-contrastive SSL.

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